Suggestion Sheet
Cloud IT Online Batch-14
Bangladesh Bank- Assistant Director (ICT)
Topics:- Server, Storage, Data Center and Cloud Computing

❖ Most Important Topics

Priority 1:- Virtualization and Data Center

- Virtual Machine architecture and working process *[BB-AME-2023]*
- Type 1 vs. Type 2 Hypervisor *[Agrani-17]*
- Server consolidation and virtualization benefits
- Data center components and architecture
- Horizontal vs. vertical scaling
- VMware and Container. *[Officer IT -2026]*
- RAID, SAN, NAS, and storage technologies. *[Officer IT -2026]*

Priority 2:- Networking and Infrastructure Services

- DNS working process and DNS record types
- DHCP working process and DORA
- System center, Patch Management.
- Active Directory architecture and functions
- Domain Controller, LDAP, Kerberos, and Group Policy

Priority 3:- Cloud Computing, System Administration and Web Technology

- A startup company wants to launch a new web application. They do not want to manage any underlying hardware, operating systems, or even the runtime environment; they only want to focus on writing and deploying their code. Based on your understanding of Cloud Service Models, which model (IaaS, PaaS, or SaaS) is most appropriate for them? Provide two real-world examples of platforms that provide this specific type of service. *[SO IT -2026]*
- How do you assessment and audit reports help detect vulnerabilities and ensure compliance to improve cloud security posture. *[BSCL -2026]*
- What is Cloud computing? Mention its service model. *[Officer IT -2026]*
- Characteristics of cloud computing. Types of clouds computing and its application. Advantage of Cloud Computing. *[SO IT -2026][NPCBL-2026]*
- What are the three types of services provided by the cloud? *[BPSC-2021]*
- WWW architecture and URL processing
- Web server vs. application server
- Load balancer, clustering, and high availability
- Backup, disaster recovery, RTO, and RPO

❖ Suggested Questions:

1. What is a hypervisor? Explain Type 1 and Type 2 hypervisors with diagrams.
2. What is server consolidation? Explain how virtualization reduces hardware and operational costs.
3. Differentiate between a Virtual Machine and a container. Explain the role of hypervisor. *[Officer IT -2026]*
4. Explain live migration of virtual machines and its benefits.
5. Differentiate between horizontal scaling and vertical scaling with diagrams.
6. Explain the major components of a data center, including power, cooling, networking, servers, and storage.
7. What is RAID? Compare RAID 0, RAID 1, RAID 5, RAID 6, and RAID 10.
8. Differentiate between SAN and NAS.
9. What is a UPS? Explain its role in data center availability.
10. What are PUE and energy-efficient cooling in a data center?
11. Explain the DNS resolution process with a diagram.
12. Explain DHCP's DORA process: Discover, Offer, Request, Acknowledge.
13. Differentiate between SMTP, POP3, and IMAP.
14. What is a DMZ? Draw a network diagram showing the Internet, firewall, DMZ, and internal network.
15. Differentiate between a firewall, IDS, IPS, proxy, and WAF.
16. Compare SAS and SATA in terms of performance, reliability, cost, and server use.
17. Explain VLAN and network segmentation with an example.
18. Explain how HTTPS establishes a secure connection using TLS.
19. Differentiate between a load balancer and a reverse proxy.
20. Explain the purpose of NAT and its different types.
21. What is Active Directory? Explain its components and working process.
22. Differentiate between a Domain Controller, DNS server, and DHCP server.
23. What is Group Policy? Explain its use in centralized administration.
24. Describe the steps that occur when a user enters a URL into a browser.
25. Differentiate between a web server and an application server.
26. What are load balancing, clustering, and high availability?
27. Differentiate between backup and disaster recovery.
28. Explain RTO and RPO with practical examples.
29. You have been hired as a Cloud Data Engineer to design a big data analytics pipeline for an e-commerce company. The company needs to:

    - Ingest millions of real-time web server logs.
    - Store and manage petabytes of raw and structured data.
    - Process data for batch and stream analytics.
    - Provide near real-time product recommendations.
    - Run machine learning models at scale.

    From the following Hadoop ecosystem components:

MapReduce, HDFS, YARN, HBase, ZooKeeper, Pig, Hive, Mahout, Chukwa, Cassandra, Avro, Oozie, Flume, Sqoop

Identify five components from the list that you would use to design this pipeline. Justify the role of each chosen component in this scenario, focusing on its functionality.

30. What is DCIM in a data center? *[BDCCL-2022]*
31. What do you mean by TIERing of a data center? Distinguish between different data center TIER standards with illustrative figure. *[BDCCL-2022]*
32. What do you mean by dynamic capacity provisioning? Why it is essential for a data center. *[BDCCL-2022]*
33. What is a DMZ in networking? *[MGMCL-2024]*